

Tools Leveraging HUMAN MACHINE INTERFACE Are The Need Of The Hour

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TinyCAD

TinyCAD is a program for drawing electrical circuit diagrams, commonly known as schematic drawings. It supports standard and custom symbol libraries as well as printed circuit board (PCB) layout programs with several netlist formats, and can also produce spice simulation netlists. It is also often used to draw one-line diagrams, block diagrams and presentation drawings.

FidoCadJ

This is a multi-platform vector-drawing program with a complete library of electronic symbols. Schematics and drawings are stored in a very compact text format. There is no netlist concept behind the drawings, and this tool offers no simulation.

Additionally, drawings can be exported in several graphic formats such as PDF. Now, the point of focus is that this tool allows great graphical flexibility and ease of use, making it a perfect tool for exchanging sketches in forums and newsgroup discussions.

PHDL

Hardware description language (HDL) is a specialised computer language used to describe the structure and behaviour of electronic circuits and, most commonly, digital logic circuits. PHDL functions as an alternative to the

mainstream graphical schematic capture tool, and is an HDL. The language is compiled into a PCB netlist, which can then be imported into a layout tool.

PHDL has revolutionised field programmable gate array designs and digital logic circuits that have been designed and captured to overcome many of the difficulties associated with the use of schematic editing tools. This tool is speculated to become the future of PCB design tools. PHDL compiler automatically supports the output of Eagle netlists and the like, and through extending a simple Java class, you can generate a netlist in practically any format required by your choice of layout tool.

PCBWeb

PCBWeb is a free software tool for computer-aided drafting applications for designing and manufacturing electronics hardware. It contains a fully-integrated component catalogue, allowing you to easily search, filter and place parts that have both a symbol and footprint. This tool is so simple to use that if you need to create a line, simply hover over a pin or existing wire segment using a click of a mouse. PCBWeb designer supports multi-sheet schematics that are especially useful for large designs.

You, as a designer, have all the freedom to create designs based on preferred wiring methodology and the tool allows you to give it a

custom net name, too. The tool also provides for basic text and shape objects to help annotate the design, and fully supports electrical and mechanical rules checking. This includes shorted nets and spacing violations that invalidate the design. It even has a shopping cart, which lets you buy components online.

Advanced HMI

This software is a marriage of Human Machine Interface (HMI) and SCADA. HMI is basically a system for visualising and operating processes, production flows, machines and plants incorporating human interface. Advanced HMI allows you to create HMIs that are not possible with other off-the-shelf packages in SCADA world. The software is based on .NET framework and uses the popular Visual Studio as its designer.

Debian 8.5 OS

Debian is a free operating system (OS), which would be the best alternative to using popular OSes like Windows and Macs of the world. This OS contains all basic programs and utilities you would want your computer to run. Debian is packed with over 43,000 packages and precompiled software, bundled up in an effortless format for easy installation. **EFY**

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